

Our supply, which includes inventory on hand, purchase obligations and prepaid supply agreements, has grown significantly due to current supply chain conditions, complexity of our products, and recent reductions in demand. At the end of fiscal year 2023, purchase obligations and prepaid supply agreements represented more than half of our total supply. We may incur inventory provisions if our inventory or supply commitments are misaligned with demand for our products.

Our demand predictions may not be correct, as we have experienced from time to time. Product transitions are complex and frequently negatively impact our revenue as we often ship both new and legacy architecture products simultaneously and we and our channel partners prepare to ship and support new products. Our architecture transitions of Data Center, Professional Visualization, and Gaming products may impair our ability to predict demand and impact our supply mix. Qualification time for new products, customers anticipating product transitions and channel partners reducing channel inventory of legacy architectures ahead of new product introductions can create reductions or volatility in our revenue. We have experienced and may in the future experience reduced demand for current generation architectures when customers anticipate transitions, and we may be unable to sell multiple product architectures at the same time for current and future architecture transitions. While we have managed prior product transitions and have previously sold multiple product architectures at the same time, these transitions are difficult and prior trends may not continue. If we are unable to execute our architectural transitions as planned for any reason, our financial results may be negatively impacted.

We sell most of our products through channel partners, who sell to distributors, retailers, and/or end customers. As a result, the decisions made by our channel partners, distributors, retailers, and in response to changing market conditions and changes in end user demand for our products have impacted and could in the future continue to impact our ability to properly forecast demand, particularly as they are based on estimates provided by various downstream parties.

If we underestimate our customers' future demand for our products, our foundry partners may not have adequate lead-time or capacity to increase production and we may not be able to obtain sufficient inventory to fill orders on a timely basis. Even if we are able to increase production levels to meet customer demand, we may not be able to do so in a cost-effective or timely manner, or our contract manufacturers may experience supply constraints. If we fail to fulfill our customers' orders on a timely basis, or at all, our customer relationships could be damaged, we could lose revenue and market share and our reputation could be harmed.

If we overestimate our customers' future demand for our products, or if customers cancel or defer orders or choose to purchase from our competitors, we may not be able to reduce our inventory or other contractual purchase commitments. In the past, we have experienced a reduction in average selling prices, including due to channel pricing programs that we have implemented and may continue to implement, as a result of our overestimation of future demand, and we may need to continue these reductions. We have had to increase prices for certain of our products as a result of our suppliers' increase in prices, and we may need to continue to do so for other products in the future. We have also written-down our inventory, incurred cancellation penalties, and recorded impairments. These impacts were amplified by our placement of non-cancellable and non-returnable purchasing terms, well in advance of our historical lead times and could be exacerbated if we need to make changes to the design of future products. The risk of these impacts has increased as our purchase obligations and prepaids have grown and become a greater portion of our total supply while our revenue has sequentially declined. All of these factors may negatively impact our gross margins and financial results.

We build technology and products for use cases and applications that may be new or may not yet exist. Examples include our Omniverse platform and third-party large language models and generative models. Our demand estimates for these use cases and applications can be incorrect and create volatility in our revenue or supply levels, and we may not be able to generate any revenue from these use cases and applications.

Challenges in estimating demand could become more pronounced or volatile in the future on both a global and regional basis. Extended lead times may occur if we experience other supply constraints caused by natural disasters, pandemics or other events, such as the COVID-19 pandemic. In addition, geopolitical tensions, such as those involving Taiwan and China, which comprise a significant portion of our revenue and where we have suppliers, contract manufacturers, and assembly partners who are critical to our supply continuity, could have a material adverse impact on us.

The use of our GPUs for other than that for which they were designed and marketed, including new and unexpected use cases, has impacted and can in the future impact demand for our products, including by leading to inconsistent spikes and drops in demand. For example, a number of years ago, our Gaming GPUs began to be used for digital currency mining, including blockchain-based platforms such as Ethereum. It is difficult for us to estimate with any reasonable degree of precision, the past or current impact of cryptocurrency mining, or forecast the future impact of cryptocurrency mining, on demand for our products. Volatility in the cryptocurrency market, including new compute technologies, price changes in cryptocurrencies, government cryptocurrency policies and regulations, new cryptocurrency standards, and changes in the method of verifying blockchain transactions, has impacted and can in the future impact cryptocurrency mining and demand for our products and can further impact our ability to estimate demand for our products. Changes to cryptocurrency standards and processes including, but not limited to, the recently implemented Ethereum 2.0 merge may decrease the usage of GPUs for Ethereum mining as well as create increased aftermarket sales of our GPUs, which could negatively impact retail prices for our GPUs and reduce demand for our new GPUs. We previously introduced Lite Hash

Rate, or LHR, GeForce GPUs with limited Ethereum mining capability and provided CMP products in an effort to address demand from gamers and direct miners to CMP. With the Ethereum 2.0 merge, NVIDIA Ampere and Ada Lovelace architectures no longer include LHR. In addition, our new products or previously sold products may be resold online or on the unauthorized “gray market,” which also makes demand forecasting difficult. Gray market products and reseller marketplaces compete with our new products and distribution channels.

Additionally, we depend on developers and other third parties to build accelerated computing applications that leverage our platforms. We also rely on third-party content providers and publishers to make their content available on our platforms such as GeForce NOW. Failure by developers to build applications that leverage our platforms, or failure by third-party content providers or publishers to make their content available on reasonable terms or at all for use by our customers or end users on our platforms, could adversely affect customer demand.

Dependency on third-party suppliers and their technology to manufacture, assemble, test, package or design our products reduces our control over product quantity and quality, manufacturing yields, development, enhancement and product delivery schedules and could harm our business.

We depend on foundries to manufacture our semiconductor wafers using their fabrication equipment and techniques. We do not assemble, test or package our products, but instead contract with independent subcontractors. We also rely on third-party software development tools to assist us in the design, simulation and verification of new products or product enhancements. The design requirements necessary to meet consumer demands for greater functionality from our products may exceed the capabilities of available software development tools. We face several risks which have adversely affected or could adversely affect our ability to meet customer demand and scale our supply chain, negatively impact longer-term demand for our products and services, and adversely affect our business operations, gross margin, revenue and/or financial results, including:

- lack of guaranteed supply of wafer, component and capacity or decommitment and potential higher wafer and component prices, from incorrectly estimating demand and failing to place orders with our suppliers with sufficient quantities or in a timely manner;
- failure by our foundries or contract manufacturers to procure raw materials or to provide adequate levels of manufacturing or test capacity for our products;
- failure by our foundries to develop, obtain or successfully implement high quality process technologies, including transitions to smaller geometry process technologies such as advanced process node technologies and memory designs needed to manufacture our products;
- limited number and geographic concentration of global suppliers, foundries, contract manufacturers, assembly and test providers, and memory manufacturers;
- loss of a supplier and additional expense and/or production delays as a result of qualifying a new foundry or subcontractor and commencing volume production or testing in the event of a loss of or a decision to add or change a supplier;
- lack of direct control over product quantity, quality and delivery schedules;
- suppliers or their suppliers failing to supply high quality products and/or making changes to their products without our qualification;
- delays in product shipments, shortages, a decrease in product quality and/or higher expenses in the event our subcontractors or foundries prioritize our competitors’ or other customers’ orders over ours;
- requirements to place orders that are not cancellable upon changes in demand or requirements to prepay for supply in advance;
- low manufacturing yields resulting from a failure in our product design or a foundry’s proprietary process technology; and
- disruptions in manufacturing, assembly and other processes due to closures related to heat waves or other natural disasters and electricity conservation efforts.

Defects in our products have caused and could cause us to incur significant expenses to remediate, which can damage our reputation and cause us to lose market share.

Our hardware and software product offerings are complex and they have in the past and may in the future contain defects or security vulnerabilities, or experience failures or unsatisfactory performance due to any number of issues in design, fabrication, packaging, materials and/or use within a system. These risks may increase as our products are introduced into new devices, markets, technologies and applications or as new versions are released. These risks further increase when we rely on partners to supply and manufacture components that are used in our products, as these

arrangements reduce our direct control over production. Although arrangements with component providers may contain provisions for product defect expense reimbursement, we generally remain responsible to the customer for warranty product defects that may occur from time to time. Some failures in our products or services have been in the past and may in the future be only discovered after a product or service has been shipped or used. Undiscovered vulnerabilities in our products or services could result in loss of data or intangible property, or expose our end customers to unscrupulous third parties who develop and deploy malicious software programs that could attack our products or services. Defects or failure of our products to perform to specifications could lead to substantial damage to the products or the product in which our device has been integrated by OEMs, ODMs, AIBs and automotive manufacturers and tier 1 automotive suppliers, and to the user of such end product. Any such defect may cause us to incur significant warranty, support and repair or replacement costs as part of a product recall or otherwise, write-off the value of related inventory, and divert the attention of our engineering and management personnel from our product development efforts to find and correct the issue. Our efforts to remedy these issues may not be timely or satisfactory to our customers. An error or defect in new products, releases, or related software drivers after commencement of commercial shipments could result in failure to achieve market acceptance, loss of design wins, temporary or permanent withdrawal from a product or market, and harm to our relationships with existing and prospective customers and partners and consumers' perceptions of our brand, which would in turn negatively impact our business operations, gross margin, revenue and/or financial results. We may be required to reimburse our customers, partners or consumers, including for costs to repair or replace products in the field or in connection with indemnification obligations, or pay fines imposed by regulatory agencies.

For example, a defect was identified in a third-party component embedded in certain Data Center products. This defect has had, and other defects may in the future have, an adverse effect on our cost and supply of components and finished goods. These costs could be significant in future periods. We recorded a net warranty liability during fiscal year 2023 primarily in connection with this defect. While we believe we have accurately recorded for warranty obligations, we may need to record additional amounts in the future if our estimate proves to be incorrect. In general, if a product liability claim regarding any of our products is brought against us, even if the alleged damage is due to the actions or inactions of a third party, such as within our supply chain, the cost of defending the claim could be significant and would divert the efforts of our technical and management personnel and harm our business. Further, our business liability insurance may be inadequate or future coverage may be unavailable on acceptable terms, which could adversely impact our financial results.

Risks Related to Our Global Operating Business

Adverse economic conditions may harm our business.

Economic and industry uncertainty or changes, including recession or slowing growth, inflation, changes or uncertainty in fiscal, monetary, or trade policy, disruptions to capital markets, currency fluctuations, higher interest rates, tighter credit, lower capital expenditures by businesses, including on IT infrastructure, increases in unemployment, labor shortages, and lower consumer confidence and spending, have in the past and/or could in the future have adverse, wide-ranging effects on our business and financial results, including:

- increased costs for wafers, components, logistics, and other supply chain expenses, which have negatively impacted our gross margin and may continue to do so;
- increased supply, employee, facilities and infrastructure costs and volatility in the financial markets, which have reduced and may in the future reduce our margins;
- decrease in demand for our products, services and technologies and those of our customers, partners or licensees;
- the inability of our suppliers to deliver on their supply commitments to us and our customers' or our licensees' inability to supply products to customers and/or end users;
- limits on our ability to forecast operating results and make business decisions;
- the insolvency of key suppliers, distributors, customers or licensing parties; reduced profitability may also cause some customers to scale back operations, exit businesses, or file for bankruptcy protection and potentially cease operations; lead to mergers, consolidations or strategic alliances among other companies, which could adversely affect our ability to compete effectively; and
- increased credit and collectability risks, higher borrowing costs or reduced availability of capital markets, reduced liquidity, adverse impacts on our suppliers, failures of counterparties including financial institutions and insurers, asset impairments, and declines in the value of our financial instruments.

International operations are a significant part of our business, which exposes us to us to risks that could harm our business.

We conduct our business and have offices worldwide. Our semiconductor wafers are manufactured, assembled, tested and packaged by third parties located outside of the United States, and we generated 69% of our revenue during fiscal

distribution operations. We have engaged with customers in China to provide alternative products not subject to the new license requirements, such as our new A800 offering.

Management of these new license and other requirements is complicated and time consuming. Our results and competitive position may be harmed if customers in China do not want to purchase our alternative product offerings, if customers purchase product from competitors, or if customers develop their own internal solution, if the USG does not grant licenses in a timely manner or denies licenses to significant customers, or if we incur significant transition costs.

COVID-19

During fiscal year 2023, we reopened our offices worldwide. We incurred incremental expenses and related in-office costs as we ramped onsite services.

Restrictions may be imposed or reinstated as the pandemic resurfaces, such as lockdown measures due to COVID-19 containment efforts in China. During fiscal year 2023, end customer sales for our products in China have been negatively impacted by lockdowns and this impact may continue if lockdowns return. COVID-19-related disruptions have created and may continue to create supply chain and logistics constraints. Challenges in estimating demand could become more pronounced or volatile in the future on both a global and regional basis.

Russia

In fiscal year 2023, we stopped direct sales to Russia and later in the year, we closed business operations in Russia. Direct sales to Russia in fiscal year 2022 were immaterial. Our revenue to partners that sell into Russia may have been negatively impacted due to the war in Ukraine.

Termination of the Arm Share Purchase Agreement

In February 2022, NVIDIA and SoftBank announced the termination of the Share Purchase Agreement whereby NVIDIA would have acquired Arm from SoftBank due to significant regulatory challenges preventing the completion of the transaction. We recorded an acquisition termination cost of \$1.35 billion in fiscal year 2023 reflecting the write-off of the prepayment provided at signing.

Fiscal Year 2023 Summary

	Year Ended		
	January 29, 2023	January 30, 2022	Change
(\$ in millions, except per share data)			
Revenue	\$ 26,974	\$ 26,914	— %
Gross margin	56.9 %	64.9 %	Down 8.0 pts
Operating expenses	\$ 11,132	\$ 7,434	Up 50%
Income from operations	\$ 4,224	\$ 10,041	Down 58%
Net income	\$ 4,368	\$ 9,752	Down 55%
Net income per diluted share	\$ 1.74	\$ 3.85	Down 55%

We specialize in markets where our computing platforms can provide tremendous acceleration for applications. These platforms incorporate processors, interconnects, software, algorithms, systems, and services to deliver unique value. Our platforms address four large markets where our expertise is critical: Data Center, Gaming, Professional Visualization, and Automotive.

Revenue for fiscal year 2023 revenue was \$26.97 billion, flat compared with a year ago.

Data Center revenue was up 41% from a year ago led by strong growth from hyperscale customers and also reflects purchases made by several CSP partners to support multi-year cloud service agreements for our new NVIDIA AI cloud service offerings and our research and development activities.

Gaming revenue was down 27% from a year ago reflecting lower sell-in to partners to help reduce channel inventory levels as global macro-economic conditions and COVID-19 related disruptions in China weighed on gaming demand.

Professional Visualization revenue was down 27% from a year ago reflecting a lower sell-in to partners to help reduce channel inventory levels.

Automotive revenue was up 60% from a year ago reflecting growth in sales of self-driving solutions, computing solutions for electric vehicle makers and strength in sales of AI cockpit solutions. The increase also included growth in automotive development arrangements.

OEM and Other revenue was down 61% from a year ago driven by notebook OEM and CMP. CMP revenue was nominal in fiscal year 2023 and \$550 million in fiscal year 2022.

Gross margin for fiscal year 2023 declined from a year ago, driven by \$2.17 billion of inventory charges largely relating to excess supply of NVIDIA Ampere architecture Gaming and Data Center products as compared to the demand expectations for these products, particularly for the expected demand in China. The inventory charges were comprised of \$1.04 billion for inventory on hand and \$1.13 billion for inventory purchase obligations in excess of our demand expectations.

Operating expenses, which included a \$1.35 billion acquisition termination charge related to the Arm transaction, were up 50% from a year ago. The increase also reflected compensation, data center infrastructure, and engineering development costs.

Cash, cash equivalents and marketable securities were \$13.30 billion.

During fiscal year 2023, we returned \$10.44 billion to shareholders in the form of share repurchases and cash dividends. As of the end of fiscal year 2023, we had \$7.23 billion remaining under our share repurchase authorization through December 2023.

Market Platform Highlights

Data Center revenue for fiscal year 2023 was \$15.01 billion, up 41% from fiscal year 2022. The strong growth in Data Center revenue was influenced by hyperscaler and cloud usage of our accelerated computing platforms and networking portfolio. In Data Center, we announced the NVIDIA Hopper GPU architecture and began ramping the first products based on the architecture, including the NVIDIA H100 Tensor Core GPU. The NVIDIA OVX server reference design launched for digital twins and other Omniverse applications. We completed two new large language models for cloud AI services — NVIDIA NeMo LLM and NVIDIA BioNeMo LLM. Additionally, we announced the NVIDIA Spectrum-4 end-to-end 400Gbps networking platform and began shipping Quantum-2 in December 2022.

Gaming revenue for fiscal year 2023 was \$9.07 billion, down 27% from fiscal year 2022. Gaming results were influenced by the rapid change in economic conditions causing excess inventory with our channel partners. We introduced pricing programs for our channel partners and started undershipping GPU supply to the partners so that we could lower inventory in the channel. As we exited fiscal year 2023, we have made meaningful progress in establishing lower inventory levels with our channel partners. In Gaming, we announced the new Ada Lovelace GPU architecture, and introduced the first products based on Ada, including the GeForce RTX 4090, RTX 4080, and RTX 4070 Ti desktop GPUs and laptop GPUs featured in over 170 laptop designs. We introduced NVIDIA DLSS 3 for over 50 games and applications. We brought GeForce RTX 4080-class performance to the GeForce NOW Ultimate membership tier.

Professional Visualization revenue for fiscal year 2023 was \$1.54 billion, down 27% from fiscal year 2022. Professional Visualization results were influenced by the rapid change in economic conditions causing excess inventory with our OEM partners. In Professional Visualization, we added new NVIDIA Ampere architecture RTX GPUs for workstations. We also announced Omniverse Avatar Cloud Engine and Omniverse Cloud and released a major update to NVIDIA Omniverse Enterprise.

Automotive revenue for fiscal year 2023 grew 60% compared to fiscal year 2022 to \$903 million. In Automotive, we started production of the NVIDIA DRIVE Orin autonomous vehicle SOC and introduced next-generation NVIDIA DRIVE Thor.

Critical Accounting Policies and Estimates

Management's discussion and analysis of financial condition and results of operations are based upon our consolidated financial statements, which have been prepared in accordance with accounting principles generally accepted in the United States, or U.S. GAAP. The preparation of these financial statements requires us to make estimates and judgments that affect the reported amounts of assets, liabilities, revenue, cost of revenue, expenses and related disclosure of contingencies. On an on-going basis, we evaluate our estimates, including those related to inventories, revenue recognition, and income taxes. We base our estimates on historical experience and on various other assumptions that are believed to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities.

We believe the following critical accounting policies affect our significant judgments and estimates used in the preparation of our consolidated financial statements. Our management has discussed the development and selection of these critical accounting policies and estimates with the Audit Committee of our Board of Directors. The Audit Committee has reviewed our disclosures relating to our critical accounting policies and estimates in this Annual Report on Form 10-K.

Inventories

Inventory cost is computed on an adjusted standard basis, which approximates actual cost on an average or first-in, first-out basis. We charge cost of sales for inventory provisions to write-down our inventory to the lower of cost or net realizable value or for obsolete or excess inventory, and for excess product purchase commitments. Most of our inventory provisions relate to excess quantities of products or components, based on our inventory levels and future product

purchase commitments compared to assumptions about future demand and market conditions, which requires management judgment.

Situations that may result in excess or obsolete inventory or excess product purchase commitments include changes in business and economic conditions, changes in market conditions, sudden and significant decreases in demand for our products, inventory obsolescence because of changing technology and customer requirements, new product introductions resulting in less demand for existing products or inconsistent spikes in demand due to unexpected end use cases, failure to estimate customer demand properly, ordering in advance of historical lead-times and the impact of changes in future demand, or increase in demand for competitive products, including competitive actions. Cancellation or deferral of customer purchase orders could result in our holding excess inventory.

The overall net effect on our gross margin from inventory provisions and sales of items previously written down was an unfavorable impact of 7.5% in fiscal year 2023 and 0.9% in fiscal year 2022. As a fabless semiconductor company, we must make commitments to purchase inventory based on forecasts of future customer demand. In doing so, we must account for our third-party manufacturers' lead times and constraints. In the past, our manufacturing lead times have been long, and in some cases, extended beyond twelve months for some products. We place non-cancellable inventory orders for certain product components in advance of our historical lead times, pay premiums and provide deposits to secure future supply and capacity. We also adjust to other market factors, such as product offerings and pricing actions by our competitors, new product transitions, and macroeconomic conditions - all of which may impact demand for our products.

Refer to the Gross Profit and Gross Margin discussion below in this Management's Discussion and Analysis for further discussion.

Revenue Recognition

We derive our revenue from product sales, including hardware and systems, license and development arrangements, software licensing, and cloud services. We determine revenue recognition through the following steps: (1) identification of the contract with a customer; (2) identification of the performance obligations in the contract; (3) determination of the transaction price; (4) allocation of the transaction price to the performance obligations in the contract (where revenue is allocated on a relative standalone selling price basis by maximizing the use of observable inputs to determine the standalone selling price for each performance obligation); and (5) recognition of revenue when, or as, we satisfy a performance obligation.

Product Sales Revenue

Revenue from product sales is recognized upon transfer of control of products to customers in an amount that reflects the consideration we expect to receive in exchange for those products. Certain products are sold with support or an extended warranty for the incorporated system, hardware, and/or software. Support and extended warranty revenue are recognized ratably over the service period, or as services are performed. Revenue is recognized net of allowances for returns, customer programs and any taxes collected from customers.

For products sold with a right of return, we record a reduction to revenue by establishing a sales return allowance for estimated product returns at the time revenue is recognized, based primarily on historical return rates. However, if product returns for a fiscal period are anticipated to exceed historical return rates, we may determine that additional sales return allowances are required to properly reflect our estimated exposure for product returns.

Our customer programs involve rebates, which are designed to serve as sales incentives to resellers of our products in various target markets, and marketing development funds, or MDFs, which represent monies paid to our partners that are earmarked for market segment development and are designed to support our partners' activities while also promoting NVIDIA products. We account for customer programs as a reduction to revenue and accrue for potential rebates and MDFs based on the amount we expect to be claimed by customers.

License and Development Arrangements

Our license and development arrangements with customers typically require significant customization of our IP components. As a result, we recognize the revenue from the license and the revenue from the development services as a single performance obligation over the period in which the development services are performed. We measure progress to completion based on actual cost incurred to date as a percentage of the estimated total cost required to complete each project. If a loss on an arrangement becomes probable during a period, we record a provision for such loss in that period.

Refer to Note 1 of the Notes to the Consolidated Financial Statements in Part IV, Item 15 of this Annual Report on Form 10-K for additional information.

Income Taxes

We recognize federal, state and foreign current tax liabilities or assets based on our estimate of taxes payable or refundable in the current fiscal year by tax jurisdiction. We recognize federal, state and foreign deferred tax assets or liabilities, as appropriate, for our estimate of future tax effects attributable to temporary differences and carryforwards;

The following table sets forth, for the periods indicated, certain items in our Consolidated Statements of Income expressed as a percentage of revenue.

	Year Ended	
	January 29, 2023	January 30, 2022
Revenue	100.0 %	100.0 %
Cost of revenue	43.1	35.1
Gross profit	56.9	64.9
Operating expenses		
Research and development	27.2	19.6
Sales, general and administrative	9.1	8.0
Acquisition termination cost	5.0	—
Total operating expenses	41.3	27.6
Income from operations	15.6	37.3
Interest income	1.0	0.1
Interest expense	(1.0)	(0.9)
Other, net	(0.1)	0.4
Other income (expense), net	(0.1)	(0.4)
Income before income tax	15.5	36.9
Income tax expense (benefit)	(0.7)	0.7
Net income	16.2 %	36.2 %

Revenue

Revenue by Reportable Segments

	Year Ended			
	January 29, 2023	January 30, 2022	\$ Change	% Change
	(\$ in millions)			
Compute & Networking	\$ 15,068	\$ 11,046	\$ 4,022	36 %
Graphics	11,906	15,868	(3,962)	(25)%
Total	\$ 26,974	\$ 26,914	\$ 60	— %

Compute & Networking - The year-on-year increase was led by growth from hyperscale customers and also reflects purchases made by several CSP partners to support multi-year cloud service agreements for our new NVIDIA AI cloud service offerings and our research and development activities. The increase was also related to the growth in Automotive. CMP contributed an insignificant amount in fiscal year 2023 compared to \$550 million in fiscal year 2022.

Graphics - The year-on-year decrease primarily reflects lower sell-in to partners to help reduce channel inventory levels as global macro-economic conditions and COVID-19 related disruptions in China weighed on gaming demand.

Concentration of Revenue

Revenue from sales to customers outside of the United States accounted for 69% and 84% of total revenue for fiscal years 2023 and 2022, respectively. The decline in revenue outside the U.S. was primarily driven by China and Taiwan related to Data Center and Gaming. Revenue by geographic region is allocated to countries based on the billed location even if the revenue may be attributable to end customers in a different location.

No customer represented 10% or more of total revenue for fiscal years 2023 and 2022.

Gross Profit and Gross Margin

Gross profit consists of total revenue, net of allowances, less cost of revenue. Cost of revenue consists primarily of the cost of semiconductors, including wafer fabrication, assembly, testing and packaging, board and device costs, manufacturing support costs, including labor and overhead associated with such purchases, final test yield fallout, inventory and warranty provisions, memory and component costs, tariffs, and shipping costs. Cost of revenue also includes acquisition-related costs, development costs for license and service arrangements, IP-related costs, and stock-based compensation related to personnel associated with manufacturing operations.

Gross margin was 56.9% and 64.9% for fiscal years 2023 and 2022, respectively. The decrease in fiscal year 2023 was primarily due to \$2.17 billion of inventory provisions in fiscal year 2023, which consists of approximately \$1.04 billion for inventory on hand and approximately \$1.13 billion for inventory purchase obligations in excess of our current demand projections.

Inventory provisions totaled \$2.17 billion and \$354 million for fiscal years 2023 and 2022, respectively. Sales of inventory that was previously written-off totaled \$137 million and \$111 million for fiscal years 2023 and 2022, respectively. As a result, the overall net effect on our gross margin was an unfavorable impact of 7.5% and 0.9% in fiscal years 2023 and 2022, respectively.

Compute & Networking - The gross margin of our Compute & Networking segment decreased during fiscal year 2023 when compared to fiscal year 2022, primarily due to inventory provisions.

Graphics - The gross margin of our Graphics segment decreased during fiscal year 2023 when compared to fiscal year 2022, primarily due to inventory and related provisions and lower margins of GeForce GPUs.

Operating Expenses

	Year Ended			
	January 29, 2023	January 30, 2022	\$ Change	% Change
(\$ in millions)				
Research and development expenses	\$ 7,339	\$ 5,268	\$ 2,071	39 %
% of revenue	27.2 %	19.6 %		
Sales, general and administrative expenses	2,440	2,166	274	13 %
% of revenue	9.1 %	8.0 %		
Acquisition termination cost	1,353	—	1,353	100 %
% of revenue	5.0 %	— %		
Total operating expenses	\$ 11,132	\$ 7,434	\$ 3,698	50 %
% of revenue	41.3 %	27.6 %		

The increase in research and development expense for fiscal year 2023 was primarily driven by increased compensation, employee growth, engineering development costs, and data center infrastructure.

The increase in sales, general and administrative expense for fiscal year 2023 was primarily driven by increased compensation and employee growth.

We recorded an acquisition termination cost related to the Arm transaction of \$1.35 billion in fiscal year 2023 reflecting the write-off of the prepayment provided at signing.

Other Income (Expense), Net

	Year Ended			
	January 29, 2023	January 30, 2022	\$ Change	% Change
(\$ in millions)				
Interest income	\$ 267	\$ 29	\$ 238	821 %
Interest expense	(262)	(236)	(26)	11 %
Other, net	(48)	107	(155)	(145)%
Other income (expense), net	\$ (43)	\$ (100)	\$ 57	(57)%

Interest income consists of interest earned on cash, cash equivalents and marketable securities. The increase in interest income was primarily due to higher yields earned on our investments.

Interest expense is primarily comprised of coupon interest and debt discount amortization related to our notes. The increase in expense reflects interest on the \$5.00 billion debt offering issued in June 2021.

Other, net, consists primarily of realized or unrealized gains and losses from investments in non-affiliated entities and the impact of changes in foreign currency rates. Change in other, net, compared to fiscal year 2022 was primarily driven by mark-to-market losses from publicly traded equity investments and changes in value from our non-affiliated private investments. Refer to Note 8 of the Notes to the Consolidated Financial Statements in Part IV, Item 15 of this Annual Report on Form 10-K for additional information regarding our investments in non-affiliated entities.

Income Taxes

We recognized income tax benefit of \$187 million for fiscal year 2023 and income tax expense of \$189 million for fiscal year 2022. Income tax as a percentage of income before income tax was a benefit of 4.5% for fiscal year 2023 and an expense of 1.9% for fiscal year 2022.

Beginning in fiscal year 2023, the 2017 Tax Cuts and Jobs Act, or TCJA, requires taxpayers to capitalize research and development expenditures and to amortize domestic expenditures over five years and foreign expenditures over fifteen years.

NVIDIA CORPORATION AND SUBSIDIARIES
NOTES TO THE CONSOLIDATED FINANCIAL STATEMENTS

Note 1 - Organization and Summary of Significant Accounting Policies

Our Company

Headquartered in Santa Clara, California, NVIDIA was incorporated in California in April 1993 and reincorporated in Delaware in April 1998.

All references to "NVIDIA," "we," "us," "our" or the "Company" mean NVIDIA Corporation and its subsidiaries.

Fiscal Year

We operate on a 52- or 53-week year, ending on the last Sunday in January. Fiscal years 2023 and 2022 were both 52-week years. Fiscal year 2021 was a 53-week year.

Reclassifications

Certain prior fiscal year balances have been reclassified to conform to the current fiscal year presentation.

Prior period intangible asset gross carrying amount and accumulated amortization in Note 7 have been adjusted to write off immaterial fully amortized intangible assets as of January 30, 2022.

Principles of Consolidation

Our consolidated financial statements include the accounts of NVIDIA Corporation and our wholly-owned subsidiaries. All intercompany balances and transactions have been eliminated in consolidation.

Use of Estimates

The preparation of financial statements in conformity with U.S. GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosures of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenue and expenses during the reporting period. Actual results could differ materially from our estimates. On an on-going basis, we evaluate our estimates, including those related to revenue recognition, cash equivalents and marketable securities, accounts receivable, inventories, income taxes, goodwill, stock-based compensation, litigation, investigation and settlement costs, restructuring and other charges, and other contingencies. The inputs into our judgments and estimates consider the economic implications of COVID-19. These estimates are based on historical facts and various other assumptions that we believe are reasonable.

In February 2023, we completed an assessment of the useful lives of our property, plant, and equipment. Based on advances in technology and usage rate, we increased the estimated useful life of a majority of the server, storage, and network equipment from three to a range of four to five years, and assembly and test equipment from five to seven years. This change in accounting estimate became effective at the beginning of fiscal year 2024. Based on the carrying amounts of a majority of our server, storage, network, and assembly and test equipment, net in use as of the end of fiscal year 2023, it is estimated this change will increase our fiscal year 2024 operating income by \$133 million as a result of the reduction in depreciation expense.

Revenue Recognition

We derive our revenue from product sales, including hardware and systems, license and development arrangements, software licensing, and cloud services. We determine revenue recognition through the following steps: (1) identification of the contract with a customer; (2) identification of the performance obligations in the contract; (3) determination of the transaction price; (4) allocation of the transaction price to the performance obligations in the contract (where revenue is allocated on a relative standalone selling price basis by maximizing the use of observable inputs to determine the standalone selling price for each performance obligation); and (5) recognition of revenue when, or as, we satisfy a performance obligation.

Product Sales Revenue

Revenue from product sales is recognized upon transfer of control of products to customers in an amount that reflects the consideration we expect to receive in exchange for those products. Certain products are sold with support or an extended warranty for the incorporated system, hardware, and/or software. Support and extended warranty revenue are recognized ratably over the service period, or as services are performed. Revenue is recognized net of allowances for returns, customer programs and any taxes collected from customers.

For products sold with a right of return, we record a reduction to revenue by establishing a sales return allowance for estimated product returns at the time revenue is recognized, based primarily on historical return rates. However, if

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NOTES TO THE CONSOLIDATED FINANCIAL STATEMENTS
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	Year Ended		
	January 29, 2023	January 30, 2022	January 31, 2021
	(In millions)		
Reconciling items included in "All Other" category:			
Stock-based compensation expense	\$ (2,710)	\$ (2,004)	\$ (1,397)
Acquisition termination cost	(1,353)	—	—
Acquisition-related and other costs	(674)	(636)	(836)
Unallocated cost of revenue and operating expenses	(595)	(399)	(357)
Restructuring costs and other	(54)	—	—
IP-related and legal settlement costs	(23)	(10)	(38)
Contributions	(2)	—	—
Total	\$ (5,411)	\$ (3,049)	\$ (2,628)

Revenue by geographic region is allocated to individual countries based on the billing location of the customer. End customer location may be different than our customer's billing location. The following table summarizes information pertaining to our revenue from customers based on the invoicing address by geographic regions:

	Year Ended		
	January 29, 2023	January 30, 2022	January 31, 2021
Revenue:	(In millions)		
United States	\$ 8,292	\$ 4,349	\$ 3,214
Taiwan	6,986	8,544	4,531
China (including Hong Kong)	5,785	7,111	3,886
Other countries	5,911	6,910	5,044
Total revenue	\$ 26,974	\$ 26,914	\$ 16,675

No customer represented 10% or more of total revenue for fiscal years 2023, 2022 and 2021.

Two customers accounted for 14% and 11% of our accounts receivable balance as of January 29, 2023. Two customers each accounted for 11% of our accounts receivable balance as of January 30, 2022.

The following table summarizes information pertaining to our revenue by each of the specialized markets we serve:

	Year Ended		
	January 29, 2023	January 30, 2022	January 31, 2021
Revenue:	(In millions)		
Data Center	\$ 15,005	\$ 10,613	\$ 6,696
Gaming	9,067	12,462	7,759
Professional Visualization	1,544	2,111	1,053
Automotive	903	566	536
OEM & Other	455	1,162	631
Total revenue	\$ 26,974	\$ 26,914	\$ 16,675

The following table presents summarized information for long-lived assets by country. Long-lived assets consist of property and equipment and exclude other assets, operating lease assets, goodwill, and intangible assets.

	January 29, 2023	January 30, 2022
Long-lived assets:	(In millions)	
United States	\$ 2,587	\$ 2,023
Taiwan	702	379
Israel	283	185
Other countries	235	191
Total long-lived assets	\$ 3,807	\$ 2,778